

Alexander Ororbia ~ Active Inference Insights 008 ~ Mortal Computation, Cybernetics, AI

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hello everyone and welcome back to active inference insights I'm your host darus P way and today I have the pleasure to chat to Alexander orobia Professor orobia is an assistant professor of computer science at Rochester Institute of Technology where he directs the neural adaptive Computing laboratory his work focuses on developing new learning procedures and computational architectures that embody various properties biological neurocircuitry and are Guided by theories of mind and brain functionality last year he published the paper mortal computation a foundation for biometric intelligence with Carl friston which grounded the Mortal computation thesis in the active inference framework in which they argued that active inference might prove useful in guiding the construction of computational systems Alex thank you so much for joining me it's been a little while since we've done one of these so I'm going to be a little bit Rusty um as exemplified by my my pronunciation of the word labor Tre uh lab I normally just go for lab um how are you I'm doing great thank you for having me on and uh no you're very welcome you can call it laboratory it sounds nice and formal I like it yeah laboratory lab um great well before we dive into that paper I wanted to ask you're kind of coming from a computational background um I've never actually asked this question first up to a guest but it struck me that I probably should which is how did you come across active inference as a as a framework oh that's that that's a nice question um well it was a few years ago and the short compressed version of that is that I was working with some uh colleagues in the Imaging science department here at RIT they do eye tracking and uh cognitive science research I also have a cognitive Science Background or part partial background in my Gra grad school years and uh so as we were uh talking about building computational models for uh ey tracking problems uh we we ended up settling on this reinforcement learning Dynamic control formulation of the particular problem we were solving and uh I had been long thinking about biological process models for reinforcement learning control and so I was already aware for a very long time of the free energy principles so I because I I'm a predictive coding

researcher I like to also pitch that as something I have quite a body of work uh build been building a body of work on and so active inference is like an extension of that right it's basically say let me put action into this process and actively sample the environment uh to facilitate my self- evidencing and so uh it just kind of made sense that we would maybe look at the free energy principle originally wanted to build predictive coding models of our reinforcement learning system and then we said well wait a minute uh exploration is going to be a difficult component here and that's what active inference brings one of the big things it brings to the table uh to use Carl's phrase uh which is this epistemic foraging I the idea that we're doing intelligent exploration driven by our surprisal and so it just kind of made sense to investigate this framework in a in a in a further further away uh and then since then it what was nice about that group is that uh my my colleagues uh weren't very familiar with uh predictive coding so I gave them a talk and then we wanted to understand active inference so we had a little reading group that we established with the grad students the other faculty and uh you know we had forced ourselves to dive a bit deeper into for example Carl's work um Chris Buckley's work as well so I was already familiar with Chris even before I actually got to meet him years later um and then one of our one of my primary grad students uh George he was uh actually building some active inference models uh and then we effectively settled on a version of that in our it tracking problem and actually that led to a really nice article about last year with George in h frontiers of computational Neuroscience where we actually published a little model uh that did very simple active inference uh in a in a in a gaze modeling task and a very simple tracking task so it was really nice and uh yeah and as then I guess to wrap up that story without uh waxing too much about my past um uh because of that uh and my fascination with predictive coding circuitry and biologic biological modeling um I sort of branched off in parallel to while George was working on that project to say um why isn't active inference even though Carl has presented it in several ways uh uh in terms of predictive coding why aren't we building things in this very very raw low-level model right because the idea is a lot of active interest use back propagation of Errors deep neural networks great idea gets you to scale it up um but uh I have a long history of uh being at odds with back propagation and uh and the non-biological ways we do deep learning and so I had a series of papers over the last few years where uh I was saying uh let's to active inference uh completely Wholesale in terms of predictive coding and so that that eventually accumulated in a a robotic control paper I believe that got published beginning of last year too uh and that was really nice because that's like completely done in terms of predictive coding um so that's I think I'll stop

there that's sort of how I found my way uh into the deep active inference corollary of the free energy principle which I've already danced over the years by saying that uh the models that I build are doing like a an approximate version of very oral free energy minimization so right right yeah yeah you bought there's a lot there um I think it's interesting yeah in in the best way possible there's a lot there in so far as I guess free the free energy principle and active inference is kind of hierarchical in of itself as a theoretical framework because yes we're minimizing our variational free energy but the way that we're doing that is kind of still up for grabs whether it's predicted processing in continuous State spaces whether it's a kind of more um message passing scheme like a pom DP scheme I think people don't realize that those debates or those theoretical Frameworks are still getting fleshed out um so it's it it makes sense why you know it is totally viable to push in One Direction maybe against back propagation and pro predictive coding for example um you also brought up reinforcement learning we I'd like to backtrack to that later on in the conversation because I'm sure you know you all know far more than I do there's kind of this live debate again still whether reinforcement learning is going to get us there um and we can talk about the developments uh in that domain but as I mentioned in my introduction really what I want to speak about is this paper uh mortal computation a foundation for biometric intelligence which I got to say I really really loved reason why I wanted to have you on is because I really really enjoyed that paper I'm coming at this from a sort of cognitive science philosophy background and I thought it was really impressive the way that you managed to integrates loads of different Frameworks from sort of pure machine learning and physics to philosophy phenomenology and cognitive science so there's going to be loads for us to talk about before we sort of dive into the details I'm sure plenty of people have heard the uh out of the how many words are there in the title seven they might be a bit confused about a couple of them so two questions feel free to take them in whatever order you wish the first is what do you mean by mortal computation and what is biomimetic intelligence okay um yeah that's a good way to start these terms might not be thrown around enough in enough Fields um so biomimetic intelligence you some might have heard of biomimicry right which is the idea we're copying uh that is that which is biotic or biological or natural uh and the idea is that we look to Natural organism maybe animals humans even very very simple life forms as you know you brought up in the paper um there's we use a lot of different examples to exemplify Concepts and the idea of biomimetic intelligence is we say well for example an animal is intelligent is able to engage in certain Behavior it is able to adapt to certain aspects of its environment it's able to forage it's able to

uh do a whole variety of cognitive tasks and um the idea of biometric intelligence is that in the Sciences of the artificial with an art AR uh what do you call it a technological artifact uh something that isn't maybe naturally we would argue biotic it is in silico uh or uh you know non-biological material can we get this object to emulate or simulate various properties and aspects of that biotic organisms uh Behavior so that is where the biomimicry comes we are sort of copying that which we have observed in nature something that is natural and trying to instill this in some mechanical computational fashion in uh a technological artifact that is not grounded in that natural biotic kind of setting so that's broadly what you could say is biomic intelligence and of course you could also take that another step further and think of artificial intelligence right in its own very word artificial means we are really building something that is essentially divorced from what we would call Natural or biological this is in the you know in silico systems that we have today uh biomedic intelligence says well maybe we should move a little towards what we would call naturalistic artificial intelligence something that draws its inspiration at the very least if not maybe Faithfully modeling because there's also different degrees of how close you get to biology in nature uh and can we construct artificial systems that are intelligent and adaptive uh that maybe for example emulate what a cat would do when it's you know in its natural environment and uh trying to for example hunt for mice or hunt for food um and then just as a quick fun word to tag on that because we also use that in the paper bionic engineering is actually the application of biomic intelligence or the uh the idea of actually formulating this and solving a real world task or problem and that can also start to bleed and it's kind of interesting part of the other fun of writing this paper is learning a little bit of the history and the origins of words and uh who defines this and everyone has kind of a different definition of these but uh bionic intelligence could kind of bleed into hybrid systems like oh human augmented systems maybe we're you know creating cyborgs or uh you know enhancing uh a natural organism in some way that they don't have with their own actual body or form so bionic intelligence is kind of bionic engineering is a little bit broader but I just like to tag that on because we also kind of abbreviate them sometimes we'll Carl and I will say biomimetics but that's just biomimetic intelligence um uh at least the way we refer to it and then bionics is bionic engineering so I think that'll stop there because we could keep going on there's a lot of history behind biomic intelligence and bionic engineering so mortal computation which is kind of at the heart and core of this paper I mean obviously it takes the entire paper to sort of scope it out I I will first pay homage and respect to my uh my friend Jeff Jeff Hinton he's actually the one that introduced that very phrase uh beginning of December of

last year if I recall correctly um in one of his papers so again I work a lot in biological credit assignment how do we train neural systems in a biologically inspired fashion and so Jeff had a paper on the Ford Ford algorithm and so I was in correspondence with him about that not too long after that came out and worked on some things of my own but he introduced in that paper in about a paragraph or two uh just talking about Mortal computation as just this kind of concept of uh hardware and the relationship to software and how those two uh you know we treat them in computer science specifically and I'm going to fold hopefully no one that might watch this might uh you know uh get angry at this but I also fold computer science with cybernetics uh because you know there's all the computational components and uh there's a lot of entanglement if you look at their history too so the ideas that we treat in many sense software as Immortal that means it can be run on any platform it doesn't really matter uh the hardware or substrate that it is instantiated on uh and again we can I we tie it in this paper to machine learning and deep learning uh which is just effectively a program that is data driven and tries to adapt based on uh input that we give to the system and uh those are no different they're like any other computer program they you could take for example chat GPT or any of the big transformers of today and if your uh you know your GPU server falls apart uh you can just take that program that you've saved on disk elsewhere and run it on another one and it will behave roughly the same um and so that's what makes it Immortal uh you can copy this anywhere you want mortal computation says well we we should consider the fact and again this is just a little bit based on and we we flesh this out a little bit more than I think Jeff at least wanted to at the time which is that mortal computation bounds or entangles the software with the hardware we we shouldn't be thinking of them as two separate and distinct entities um and so that's where the phrase sort of originates and one of the core motivations uh is that um we get a certain Energy Efficiency um because one of the biggest criticisms of uh Transformers and neural models today that characterize artificial intelligence is that they are they greatly energy inefficient uh they they require a lot of compute uh and while the costs are going down and I'm seeing some of those Trends pop up uh they they leave a pretty big carbon footprint um and so that's there's also a fun and another vein of research that got into this paper of my own is I've been studying green AI versus red AI uh and you know just like we talk about green initiatives because we you know care about uh global climate change and our concerns of what's Happening to our planet and our environment um that's green AI we would want to build intelligence systems that don't leave such a big carbon footprint uh red AI is what we currently have and there was some really nice studies Emma strubel many many years ago put

out one of the first uh measurements of these language large language models um and so mortal computation uh is one way to address this great inefficiency um Beyond a lot of other things but that's kind of where at least Jeff uh was coming at this from um what kind of funny just as a quick side note uh because I obviously track Jeff's work and uh well he in December you know sort of went down the route of you know we should you know get rid of backpropop we should go to some biological form of learning and here's my really wonderful idea uh and Mortal computation sort of one of those goals uh later he sort of I noticed used that as an interesting counterargument to talk about the dangers of immortal computation as he's talking today about you know the potential threats uh that Immortal systems like chat GPT and Transformers like them and other models not just these large language models posed to humanity and it's bringing up an interesting discussion and thoughts about you know an existential crisis for Humanity and worthy things to be talking about but he sort of steered in the direction I noticed he uses mortal computation sort of as like a well that's what you know we would say maybe the brain and you know biological organisms do uh but the models that we have today don't do that and actually that gives them sort of like superpowers and some dangerous abilities that we can't do with humans and that's worthy of discussion but we obviously don't go into that in this paper um so that's where it sort of originated and then just to sort of quickly cap it off because I know you said you want to talk more about the paper because the whole paper goes into What would mean to be mortal computation uh I also really liked in one of the talks that my friend Carl gave or had with someone else I was interviewing him about this work too and he said that uh this paper's form of mortal computation is sort of like another take on the concept so again like what I explained was sort of what Jeff meant by it um and then that was his launching point for a set of other arguments um this one says well wait a minute that entanglement can we characterize that and dig into that a lot deeper and so the idea is that that gets us to the origins of life right of how we ourselves you and me Darius are mortal computers in effect right right if we don't mind calling ourselves that and you know not trying to remove our humanity and so that's what sort of this survey paper ends up becoming is it says well okay if we wanted to really go down that rabbit hole if you will what does it mean to Define a mortal computer and do other domains of science have possible definitions and it that's what the other fun thing about this paper I know you didn't ask about its motivations but what's nice about it is that it is an echo of a lot of already great ideas by people way smarter than myself uh over the last century uh talking about essentially what you could call Immortal computation ranging from biophysics and biology to physiology uh to

even cognitive science has already some wonderful theoretical Frameworks that we sort of just sharpened or highlighted or there's a little section in there where Carl and I extend one of the cognitive theories maybe we'll talk about that later I like that you know the we have it called the 5e cognitive theory yes yeah yeah yeah cool yeah that's fantastic um okay perfect so there's yeah again Lots I want to unpick there so before we fold in into inactive inference more technically and talk about how that is a framework uh in which we can ground mortal computation I want to ask a question about the motivations about this Energy Efficiency um because people might be listening and saying well Immortal computation sounds kind of nice um it's kind of there and I can take it and it's substrate independent and so um my commitments are fewer why why does it make more sense to turn to a different form of computation rather than just refining um something which clearly works and uh may be uh in the long run it seems to me at least more efficient in the sense that it's just more transferable across different domains um yeah that that in that we we mentioned that a little bit in the beginning of our paper why would we want to commit ourselves to Mortal computation it's rather brittle as you pointed out so um so again there's many different reasons for why we would want a mortal computer but you said the Energy Efficiency so to let's just pick at that little bit of the argument so near the start of the paper we we talk about the thermodynamic motivations for uh mortal computers and there's some really nice ideas from physics and information Theory I mean there's uh even probably more that we don't properly scope uh and two of them that we can just briefly mention are the land hour limit and jar zinski inequality um when you in a nutshell without getting technical about them because we do I actually made a point to flesh out those equations just so readers would be like if they're not familiar they can look at them but really at the end of the day those two concepts say there is a uh a bound on the amount of work or the thermodynamic thermodynamic cost that A system can have to write and store information so again in a computer um you know the idea is that we're using bits to you know store things in memory um and so basically what those two say is that uh there's a point where we're not going to get any more thermodynamic efficiently about you know the cost of actually racing a bit I believe that's what the landar limit or sorry yeah the landar limit and J inequality partially touch on they also talk about random fluctuations and averaging things to get a system to an equilibrium state um but at the end of the day it's all just about that there's going to be a a lower bound that we can't breach any further and what does that actually imply for computer science actually computer scientists have written about uh these physical limits uh and the thermodynamic costs and it says that it might mean that we

want to reach something called in-memory computation so in-memory Computing basically just says that we want to conduct the mathematical computational calculations let's those that characterize neuronal simulations directly at the chip right so the idea is that we have processing and we have it organized let's say in a crossbar in a neuromorphic chip and we use that as like one of our little examples and we can actually essentially construct the neural model in that memory crossbar and so like the neurons are arranged in a particular way physically actually in this little chip and the synapses relate you know are related to for example memory ERS right and the ideas that we transmit information across these and that will be at the lower limit or the the bound on computation why is this different than what we do today well today uh Transformers and the like regardless of the value of gpus and how and even what what are they call tpus tensor processing units uh and the the speed that they have afforded us and that's actually partially one of the core reasons of the deep learning Revolution uh is that we use Von Neumann architecture so the idea in that is that there's a greater cost Associated just fundamentally by simulating these neural models on a von Neumann architecture and that's because you could look at for example the memory organization of a computer so uh Beyond you know like let's just call it slow long-term memory systems which you know you can write and this will preserve your information when you turn off your computer and turn it on later we kind of want that uh versus volatile memory which is something that is really fast doesn't give you much of a cost to store things but it's volatile so obviously if you turn things off that you can't be guaranteed that your information is stored on there and then there's different types of intermediate memory and so the way of memory architecture sort of is structured is it arranges this memory in a hierarchical fashion and the CPU lives or the GPU if you will but let's just say the CPU specifically lives on top of this memory hierarchy that's where your computer programming lives that's where our neural network is being executed by the compiler which talks to the assembler and it talks to the bits at the very low level of the computer and so that's where your mathematical model is but it needs data it needs information it needs the matrices or tensors that store the synaptic weights right let's just think of our traditional neural network it's just a bunch of Matrix multiplications and some nonlinearities well you need to retrieve that weight Matrix for the first layer well that lives at the very very bottom of this hierarchy so there needs to be some transfer process and that's already exhibiting a great thermodynamic cost they've Quantified that I remember putting in the paper like certain uh measurements of energy at different layers so already fundamentally we are not operating as

close to the lower limit on thermodynamic cost as we could be um and so that's already right there you that explains a great deal of the thermodynamic inefficiency of deep neural networks and intelligence systems that we have today um one of the motivations you can use the idea of in-memory compute uh in-memory compute uh is that it gets you to something like Edge Computing so if you have something in robotics which I'm very fascinated by robotic systems or even autonomous vehicles um you're not going to be able to transmit to servers very easily right you're not going to be offloading your intelligence somewhere else uh because then you have the communication cost to pay of transmitting information you're going to want your system to live in the device that you're using so in a robotic system you have limited power you have limited uh compute resources so you're going to want to work with in-memory Computing like for example a great uh in-memory system would be like a neuromorphic chip just in case the audience is like well what would be examples of that um and one that has fascinated me which is where the realm of spiking neuronet comes into play and I have done research in that as well and that's what gets you to neuromorphic chips which does rest on this principle of in-memory computing right so the the actual spiking neuronal models are actually laid out in these neuromorphic chips and computation happens directly on there so um I think that's sort of like the uh the in energy kind of argument for why we would want this this would be quite useful for Edge Computing cases and you you'd say well okay well that's just a few examples right right we're doing a lot of intelligence without that but most of Robotics is going to be Edge Computing if we want an intelligent uh humanoid or non-humanoid like robotic system um we're not going to be running them on big supercomputers or gpus and again I'm not trying here to attack those that maybe are for like hive mind intelligence where we do communication across servers and we can pay that cost um but if we want these organic like robotic systems this is where I think mortal computation really uh is a path forward for those types of applications and there's a lot of other applications that you could argue have Edge Computing so um I think I'll stop there because that just centers only on the thermodynamic argument um I'll just quickly before I forget it tag on uh the other part of is that when we get to that lower limit on thermodynamic cost uh there's a nice equivalence to the variational free energy minimization process because uh while again I I did tend to focus a little bit more on the thermodynamics the physics side of things you know I liked how Carl and I sort of connected it to the information theoretic idea that uh to be Baye optimal right we want to actually get close to that limit and so there's it's sort of like there it's saying the same thing actually the nice thing about the variational free energy princip sorry process is it is also

actually optimizing that thermodynamic cost in organic systems so we are operating at that lower limit that's what it means to be Bay's optimal free energy perspective and I'll stop there but I like that I just want to make that connection because that kind of tags a little bit in that argument and then it starts to bleed into the other motivations you could say for moral computation yeah yeah yeah no I let's actually stick on that because I think that's a really interesting point um I think people some you know it's it's actually possible to kind of do the free energy principle without taking into account thermodynamics right you can just do the statistical mechanics and the information Theory just do Shannon information and just say you know we're reducing self information so people might be curious actually how this ties to thermodynamics I'm curious about how ties to thermodynamics I'm not a physicist so to minimize variation on free energy is to be sort of operating at this um lower Bound in in terms of thermodynamic energy what does that look like and I think you talk in your paper about kind of exchanging um energy or or or the work needed to maintain yourself at this bound um what what what does that look like in terms of the mapping between information theoretic terms and thermodynamic um terms well first of all I'm not a physicist either so we we should be clear here as well I am a computer scientist and cognitive scientist um so again it goes back to like what I and I mean I have my paper here just to remind myself if I need to but like I said it goes to the jain INE equality and the lar limit um so I guess maybe I want to ask a little bit of a clarification what do you mean what does it look like just to help me out here to trigger that I'm not so just in terms of that kind of isomorphism um that minimizing variational free energy just is operating at this thermodynamic lower bound MH just in very it can be very simple because it's going to have to be very simple for me um because they don't teach you this stuff in cognitive science enough yeah um what does I mean what does that mean um in terms of our sort of thermodynamic exchanges with the environment because you know I think a stumbling block for a lot of people when they hear about free energy is they there are lots of ways of talking about energy so it's kind of being able to do the mapping between all the different aspects whether we're talking about kind of Shannon information or whether we're talking about thermodynamic energy um or we're talking about whatever whatever you know whatever framework you want to come uh you want to work with so just in very simple terms what does the minimization of I mean can put put it very generally what does the minimization of variational free energy Bey Beyond moral computation Beyond computers very simply what does the minimization minimization of variational en look like in terms of bound well I guess it gets a little confusing because now I'm thinking about the actual implementation so

again I could approach it from a predictive coding point of view because that's what I usually appeal to when I think of minimizing variational free energy so I guess at the end of the day it kind of goes into I guess the problem with your question I think it makes me think of the other arguments we get into metabolic efficiency and the idea of resources that the organism is acquiring uh okay um I think I know where I want to go with this so uh it's actually kind of later in the paper where we touch back on the free energy principle and we get into this concept so what does this look like it looks like the concept of the what do you call it the non-equilibrium steady state the Ness so what does this mean more generally for even artificial and biological organisms the idea is that we want to reach a point that we are far from thermodynamic equilibrium and so yeah I gotta charge my memory here a little bit of certain parts of the paper so apologize for the delay there um so the second law of Thermodynamics is that you know that entropy must increase right in in a closed system and so we are tending towards higher entropy and we are reaching you know systems will reach equilibrium and actually that's not what living systems are doing living systems live at what I just said earlier non-equilibrium and you'd say well wait a minute are we trying to adhere to the second law of Thermodynamics and this is part of what was interesting about this paper in in understanding the history of things there is an interpretation of thermodynamics and biology and they call it the first law of thermody First Law of biological thermodynamics at least this is what I had found in my uh you know uh through the in through the literature and the idea is that organisms do want to stand far away from equilibrium because equilibrium means death equilibrium means dissolution or disintegration into a heat bath into your environment uh your identity ceases to exist and that allows me to poke and touch another concept of why mortal computation is so important it's a process of identity Constitution and preservation and so uh where that connection comes into into play is that organisms they have a goal and according to the free energy principle organisms have this this the idea that their Dynamics wants to make them stand away from this non equil sorry from this equilibrium and so they're going to need energy and resources to allow themselves to do that and then there's another part to that and actually I do want to remind myself so I don't steer the audience wrong because I liked how I pitched it or how I put it in the paper because there are two principles that before you got into the markof blanket discussion oh no h okay I remember what yeah so again it does connect to the markup blanket but the idea is that these Dynamics in the S the internal states of the organism are conditionally independent of their environment the external States right and that's what the markup blanket formalism tries to formalize that more strongly and the that we

have those density Dynamics they are conditionally independent however you are still uh weakly coupled to the environment through the maral blanket interface and so again I'm apologize for introducing those words later in case you wanted me to flesh those out but uh the idea is that because we are coupled to our environment that also gets us to the connections of active inference and why we must act to move about our environment so as to prevent to keep ourselves at the non-equilibrium steady state and the free energy principle says that uh if you are a living organism or an entity you will return to this NZ and your action in your uh your goals and behavior lead you back to the NES the other concept is and this is the part two of the free energy principle is the concept of continual self-evaluative model which is connects to the other component of the paper where we brought in cybernetics and like the good regulator theorem which we could talk about that later if you want um so I think that's what it looks like right it manifests itself in this this continuous goal of trying to stay in non-equilibrium you're avoiding equilibrium which is dissolution into your environment death which also means the end of your identity um and so that's what a mortal computer is constantly engage in and that's what the thermodynamics kind of gets you to it walks you to that which is another way that the free energy principle states that right and so you can look at it from the like the information theoretic quantity of like oh I'm you know dealing with gnats or bits right but the idea is that the end of the day you trying to stay in your nest n steady state and you're continuously self-evidence say uh and you're dealing with this weekly coupled weak coupling to your environment and you're interacting through your markup blanket I'll leave that black box for now in case you want to poke at no that that that's great I mean that that's much more my language ense my that's what I think it looks like that's where the thermodynamics really gets you into the biological thermodynamics the first law biological which is really what that is sort of restating REM manipulating it's basically all talking about that non-equilibrium that we want to live at which then you can and the nice thing about the biological first law of thermodynamics is it sort of deals with the fact that you're saying well we don't violate the second law of Thermodynamics entropy will still go up it's just not in the organism system itself it's open system that lives in a close system which is a greater like in econ Niche organism pairing right and so entropy will go up according to the second law in the entire system but inside this system this explains why entropy can go down and that's what you know and that's what I liked about that because that was a great source of confusion I saw even in the literature um you know how are we dealing with the actual second law of Thermodynamics well we're not Viola we are really adhering to the biological first law of

thermodynamics which that's what NES and the free energy principle sort of at least one dimension or the dimensions that I'm more familiar with tackle yeah so that great so that latter point is kind of where I was heading with my rubbish physics vocabulary but the the nest and the TR to set stuff is is is all perfect for us so don't worry you haven't opened Pandora's Box because this is a language at least that I'm speaking these days and and this institute is speaking but that's the point I was making um because it's it's kind of all about the the scope of one's analysis right like equilibrium for whom um because I think people get confused that they they think of free energy uh minimization as homeostasis which they learned at school as equilibrium but then they're saying that we're far from equilibrium um so it's it's it's non-equilibrium with respect to the universe or with respect to the way the universe wants to be which is dissipative fundamentally right but it's it's equilibrium for ourselves in terms of our internal Dynamics given our steady state you know given our attractor set and I should add a word that just complements what you said because you're correct homeostasis is actually kind of confusing and we introduced another word uh and I thinkk Carl for pointing me in this direction because he he had us you know highlight this hormesis I think yes because it handles this and we could talk about that if we want but I that's the word people learn but I never learned that when I was much younger I I only learned that in the context of formulating this framework with Carl so yeah and then people uh I guess there's a great deal of emphasis on allostasis as well within active influence which we can touch on we we should probably unpick those three yeah all thees and don't forget autopoesis as well autopoesis I've been reading a lot about autopoesis um great okay cool yeah I I mean should I keep perusing down this area I think it's interesting because it's like so we're minimizing entropy within our own bound within our own system whatever that looks like because those boundaries it's it's hard to kind of cast a limit but that according to the second law of thermodynamics means that we have to be creating entropy elsewhere just so that the system itself is abiding by the second law um what how do we do that how does anything do that where where are we creating entropy creating ah okay well let me make sure that I zoom in on the right component that touches on that so I think the problem is there's all these slices to Mortal computation as you're familiar with the paper and I think a piece of one of the other interpretations is a more approachable way to understand how we increase entropy which is the idea of inactivism so this is I'm I'm borrowing from so just for the audience that maybe hasn't read the paper uh we intro we we sort of have a pathway that we talk about the Mortal computation thesis we talk about biophysics we talk about the cybernetics interpretation we talk about the cognitive science uh

framing right even though they don't use these words they're essentially saying these ideas and then we lead into sort of a free energy definition and a Markov blanket formulation with uh other things and then we lead to a definition so that's so when I say slices that third of that one portion of the paper on cognitive science is where we borrow uh from uh well what's normally called 4E cognitive theory which you know and one of those e is inactive the first one's embodiment which we can touch on that one as well but uh inactive is about your relationship to your environment and so uh and how that interacts with you there's like a bidirectional relationship we act on our environment and that effectively means we are authors of the information uh that we will end up using and uh the resources that come into us right in terms of our foraging in terms of our survival remember not to forget our goal is to stay in non equilibrium right the non equilibrium steady state and so uh That's essential how we're going to end up creating entropy right because we are going to manipulate the environment the idea where the active portion the active inference element comes into play nice and so we are uh what's the right word that we use in the in the paper that others have used as well uh we are engaged in Niche construction right indeed yeah so because we manipulate the environment using the affordances that it brings us and we do things to allow ourselves to again you could Tye this for those that maybe don't like Ness as the goal but say we want to survive survival and identity con continuance right is essentially living in this non-equilibrium steady state uh we enact our environment we change it and then that information of course we're then Gathering that and so the environment affects us right it's a bidirectional relation the environment uh constrains us puts uh problems on us we have to deal with that but we change the environment so we are creating disorder and that's I think key element is that through the actions that you take through any animal that that they take uh through any organism that they do to manipulate and change something about their environment so that way they could continue to get resources uh matter and energy uh those are the two the fuels of the work Universe right uh we we essentially create that disorder right and so that's how the second law of Thermodynamics get satisfied because that's what biological or biotic entities and I'll just zoom in on them for now because you know again I want to be careful not to step on physicist Toes or biophysicist toes as well because I'm not in those domains I've just read enough of that work to synthesize right um but that's how you're creating your entropy is that you are inactive right and so this is the concept of inactive uh cognition you said you're reading about autopoesis there's also this thing called what is it uh autopoeitic inactiv M as well yeah Evan Thompson yeah exactly yeah I I've just I'm rereading mind and life which is Evan Thompson's

kind of massive magnum opus from 2007 which is which is cracking um I like pointing people in directions of readings that they can do um so yeah that's a great one Thompson's someone everyone should read I think yeah yeah so yeah beautiful stuff um obviously verel rush and Thompson 91 the embody mind and all the stuff that did with matano 1980 Al pus all wonderful work yeah in mind in life he talks about the fact that inactive uh entities are uh what does he say he says operationally closed um but sense making and by sense making I think what he means is this kind of yeah thermodynamically open um so we're in the game of autoparesis but that that doesn't deny that doesn't negate the possibility of obviously of interacting with the environment in fact we have to integrate the environment into our model or into our uh action policies to use an active inference term in order to Eng like do autopesis you're not going to last very long if you're an entirely closed off system um it's funny because again it just makes me think about how scale how scale is really a very important kind of currency in this discussion and it might just be philosophically interesting but I'm sure it's interesting for physicist to too mhm because it makes me think that like whatever we are whatever the bounds of us as an identity is we're kind of pockets of the universe where entropy is decreasing but then there's nothing kind of a prior stopping that process expanding to let's say Societies or worlds right or galaxies or milky ways and and I guess the question is is uh where where does the where does what's the what's the overarching clo system uhu because you know I again I know you're this is this might not be your the question that you want to answer I don't mind waxing with your thinking you know speculating if well just seems to me that and again this is not an argument from design I think interestingly there seems to be a sort of re Resurgence perhaps of an argument from design I know Philip gfer just released his book why which is um a kind of philosophical argument for I wouldn't say it's design per se maybe it is from a S probabilistic perspective um I think relying on actually I won't I won't say because I haven't had Philip on but um you know the universe has it's it has its space at least for entropy reduction but we still have to we still have the second lamic thermodynamics operating in these close systems I guess the question I would ask to a physicist is what what if it's just open systems all the way up well yeah I would love to have be fun to pair pair up with a physicist here and ask literally that question um all I will say in my non- physicist hat uh is that at least when I was constructing this framework with Carl at the and it'd be fun to get Carl's thought on that as well but oh well you know future yeah future call in the future but uh uh I I like to think that there is some outer close system and so I guess when I was writing this in a perhaps ignorant way I was thinking of the entirety of the universe as the entire system

right that is the close system because um I thought you were gonna ask a different question and I I might I have an answer for that one but uh which go ahead about Cosmic bodies and all the other entities this universe right um uh but but just to stick to your your your comment um I think the entire universe is the close system and that's where the entropy is going up because you know you probably have heard about uh at least it's I believe it's still like more of a physics theory but the eventual heat death of the universe right everything indeed increasing heat increasing energy uh Asim Isaac Asimov has written wonderful stories about it right um and so uh I thought of the entirety of the universe and every other entity that is in it some of them could be you know living in that non-equilibrium steady state but everything eventually is tending towards the heat death of the universe so the closed system there is an outerbound I just haven't picked it specifically and that's where a physicist would be much more qualified to draw that boundary the tiny question that I was wondering because it set you were almost walked yourself there which was when you brought up bodies and I'll even throw an electrons in here too because uh I thought you were going to say our Cosmic bodies mortal computers our our electrons mortal Compu um and I do have an answer for that one it's not again I'm not going to pretend that I have the physicist answer but I think Carl has actually eloquently addressed this again in other discussions related to this paper because someone did ask him well is the moon cognitive right yeah they would also you could argue following certain Dynamics as well why didn't we include other objects because we have actually examples in the paper very specific like these entities serve as wonderful examples of mortal computers we can chat about them if you like um but I like how Carl said why does the moon not necessarily think or at least we wouldn't call it cognitive and he talks about these Dynamics the the flow right and you can decompose flow in physics and again I'll put it in the lightest interpret so that way I don't maybe misspeak um but you can decompose uh Dynamics according to helmoltz decomposition so there is something called solenoidal flow or conservative flow and then there's this thing called dissipative flow or the random fluctuations as Carl has mentioned and there the idea is that uh random fluctuations uh that's the stochasticity of any system that you must deal with and when you get close to Quantum which I am not a Quantum person person but I understand that the random fluctuations are so high the amplitude of them that they dominate the Dynamics right and so that's why in quantum mechanics you specify things in terms of probability distributions um but then as you move all the way up in increas and and Carl's also been very careful to avoid saying space which is constant but rather just the size of the objects that we're consider as we get up to uh macroscop miso macro and cosmological

scale because you mentioned scales earlier um the idea is that the flow becomes more dominated actually almost entirely if not completely solenoidal right which is this conservative Flow It's Time dependent flow dependent on the states because again I should mention to when we're talking about flow talking about the states of a system right and they change with time everything is dynamic and evolving um and so uh the dissipative random fluctuations get averaged out because as you get bigger and bigger in scale you average out fluctuations and the amplitude goes down right so the as you progressively get bigger you progressively average out the noise to the point where when you get to the cosmic bodies oh going on in solenoidal flow whereas a Quantum or the very very M the small scales it's mostly dissipative random fluctuations so how does that connect back to Mortal computation or cognition or sentience or anything of that form well Carl has used the phrase the Goldilocks regime yeah so and we didn't write about it in the paper but I thought it was a nice it's a good thing to think about that there is this regime of being big enough but not too big and not too small you know that's the goal the regime right such that you have a balance or you have a combination of dissipative flow and solenoidal flow because the idea is that you need that dissipative flow in order to minimize your variational free energy the idea is that we need that Dynamics we need something to be surprised to uh Drive our continual self-id but you need that solenoidal flow which by the solenoidal flow can be like rotational flow very like a Carl used another example just because I I remember seeing this and it's stuck in my mind about water then flowing towards a drain or a or an open plug and as it rotates around it doesn't go down that would be the dissipative part it's going down the gradient just the circular rotational movement right so that's what you know these bodies at the cosmic scale they collapse to Newtonian or Iran mechanics or classical mechanics right and the ideaas that they're just following this conservative flow but there's no dissipative random fluctuations whereas electrons and Quantum objects as well right they're just dealing with all this dissipative extreme noise or stochasticity so electrons don't think uh and the moon does not think because again you need sort of this balance between these two Flows In order to uh essentially uh promote this self-evident in a structured way too because again if everything is just so completely random uh it's you know obviously hard to form you know uh what is it steady patterns and you oh and the other component to this is the cycle and that's where I think the Mortal part comes into play the solenoidal flow or the rotational or the predictable patterns or the conservative patterns that's where you get things that repeat heat or a periodic and so that's what gets you the life cycle so in there's a lot of wonderful examples like breathing is actually

an example of that solenoidal flow or Cycles in language or the fact that you and I die right and there's the cycle of death and rebirth um those Cycles are more of that conservative flow uh and that's the part what makes creates this mortality right and that mortality the other take by the way I sort of skipping around a bit the other motivation for Mortal computation is that that cycle that life cycle adds a pressure to these entities right that would not normally be there because again if you're Immortal and you don't have to worry about death you might act a different way right so that that life cycle thing comes from that other aspect of the flow but again when you get but you need that sort of stochasticity right to you know engender that continual self- evidencing that wouldn't be happening like the cosmic scale which is just essentially all those random fluctuations are averaged out so I think uh you didn't ask that question per se but you almost and I thought you were GNA yeah thought I should answered that because I think that's something that's not really addressed in the paper and I was surprised to see a couple different places that that question comes up so no electrons don't think and the moon does not think at least in accordance to the way we were thinking about Mortal computation there is possibly a Goldilocks regime but what are the bounds of those regime I I I'm not sure yeah yeah yeah no I in in the first episode of this podcast Carlo and I go quite deep into that oh see on that already okay yeah well it it goes It goes pretty you know we start talking about information geometry spaces and um and indeed these flows yeah yeah so so just for everyone um Co does a big exposition of that in the first episode and and yeah it's beautiful because it ties together as well like this what I've been speaking about with phenomenologists especially people interested in haiger this kind of future orientated nature that we have because we're always at risk of being dispersed so we always have to be acting for our self organization not only at time T but also for t plus one yeah which which feeds into this kind of notion of AI estasis so we've we've done we've done a huge Cosmic overview without even talking about the paper but I do want to ask one more question about Immortal computation going all the way back to what versus Mortal computation going all the way back to what we were talking about at the beginning so the the way that Jeff so Jeff Jeffrey Hinson super important guy in machine learning um and I know him and Carl and P Diane there's you know they were working together in the 90s so this is a very long history Gatsby Union right yeah yeah yeah at the Gatsby exactly um and you know this casting of immortal versus Mortal as being a software Hardware dichotomy so in Mortal computation the software is kind of bound to the hardware like us right at least the way that we would uh you know most materialists would think of cognition and the Brain whereas software is um substrate

independent in the sense that you can take something like chat gpt and put on my computer put it on your computer um so that in my mind again because I'm coming at this from a philosophical perspective makes me think of functionalism as a kind of philosophical theory of mind which at least in its kind of very raw form is substrate independent and an argument which comes from Ned Block which is a kind of argument add obso is that um if you take that idea of substrate Independence seriously what you might end up with is what he called the Chinese brain which is let's replace every let's have a let's have a system where every neuron in the brain is replaced with a person from China because the number of neurons in a certain brain region you know map onto the number of people in China and if they do all the flagging right if they do like the synaptic integration then we would have to be compelled if it was completely isomorphic to say that that brain was um was conscious and he goes well this is just clearly wrong and now you have you do have functionalist who will say well actually that brain would be conscious sorry that Chinese Nation would be conscious I guess I say all of this to ask you know just purely on intuition does the notion of a kind of boundlessly immortal comput uh computer in or or or software really does that really make sense because it seems to me that yes you can access TR gbt and I can access track gbt but that is contingent on the fact that you and I are working on at least relatively similar Hardwares MH um so I'm wonder I'm just wondering whether the notion of immortal software is itself a slight red herring or an overshoot possibly um I mean I I guess I guess the way to think about what the argument for Immortal computation really says right is that um so I gotta think this one through because yeah I've not heard about you know the the actual problems with Immortal computation per se so are are you are you referring to maybe perhaps the limitations of it well just I I'm just kind of bringing into question the very you know the very dichotomy that is supposedly at the heart of the problem which is that you you have hardware systems that are uh prey to dissipative forces whether it's the metal in my laptop or the Silicon or whatever it is and then you have the software which which arguably is substrate independent right that mean that's the very dichotomy that's set up my point is is it the in the substrate Independence is is a bit of a misnomer it's not necessarily the case because all the all the evidence we have for substrate Independence are relying on very similar substrates right like we all have very similar I see you're saying that because at the end of the day we are effectively using the same type of substrate right you and I executing this so the fact that we aren't we aren't really testing the the independence component right running it on an entirely different substrate write it on a write it on the population of China and see right like you said create the uh chin chese room style the

system right could we that could we construct that um possibly I mean it is possible that it is a misnomer um I guess maybe what people think about and again you could probably even find minor criticisms or minor issues with this is like when you change the particulars of the hardware so you know when we run for example chat gpt or Transformer we're using something like a GPU System right you know maybe multiple GPUs well I could say well wouldn't changing the hardware mean I run it on a set of let's say I have a lot of CPUs lying around right that is different Hardware they have similar properties like let's say physical properties in silico right um but they are different constructs to run you know the same thing and the idea is that um you know Chat gpt would run just fine on a set of CPUs to my knowledge I haven't necessarily tested it but you can run multi CPU systems same thing with a TPU they might just have different speeds right they give you maybe uh the speed of training the convergence properties might change but at the same time they will again if we're not doing learning I would imagine the program runs the same way and I would even imagine that you can run let's say you had a really really good like device that isn't a multi-gpu system you pre-trained your model and now you run it on there it'll behave roughly the same however I could the argument that even then there's going to be differences right because there the the properties of just the actual system that you're simulating on you can even talk about for example seed generation so you know when we write a program we have a computer seed right you control the do the noise or you control the level of determinism in your computer simulation and uh you can use wall clock time for example to generate seeds well if you change the system well you're not going to have access to the exact properties that instantiated that seed on my computer so already your computer's on a different stochastic trajectory if it let's say uses noise I mean maybe Transformers might not be the ultimate best example but any system that's like a noisy neural model would already be behaving a little bit differently right you'd maybe on average they would behave the same if we averaged out those fluctuations um so there is a little bit of difference across Hardware platforms it could possibly be a misnomer and again to use your example of the uh the Chinese room style neurons and we simulate GPT right but with these particular with humans serving as those neurons um yeah I would I I don't know if I can really comment on whether or not he would that could perhaps be a way to break the uh complete immortality of the system um I think what moral computation just to make sure that I uh don't get lost in the plot here is that mortal computation really really wants to emphasize that uh at the end of the day well the way that we design our programs they really are Eternal I mean they might not necessarily run exactly

the same like you're pointing out there there could be some problems but we are not designing our intelligence systems with any uh or very little uh what do you call it constraint or informed informative information about the substrate it's going to run like when I write my neural simulation I run it on a server maybe I run it on my desktop I don't think about oh is this device actually going to affect the characteristics of my program like that's sort of what Mortal computation even touches on which is like the particular quirks of an entity are very substr it's like almost like once you build one particular neuro robot you would also get what in cognitive science there's this term coming up cognitive diversity right so if I built a different neuro robot but let's say I it has very similar structure right you would say okay well these two mortal computers I I tried to design them exactly the same I run the same program well because they are two separate entities and the way they interact with their environment they are going to have their own eccentricities their own quirks their own particular Knowledge from self-evidencing their particular selves right so the idea is that this neur robot this neur robot while they have very similar Hardware style they're still going to be two very different systems whereas Chad GPT which was let's say or Transformer let's not pick on Chad GPT but any of these big neural models that was let's say pre-trained and then you know Consolidated and said I'm going to clone you into these systems maybe their inference system right their inferences might lead them to uncover different data but at the end of the day the particularities and quirks of the software came they're kind of tied to the same origin point right so um I think that amount of adapt ity in the essence of how we again find ourselves towards the NES the non-equilibrium steady state is really dictated by that particular knowledge and information that's encoded in your body and so that's where we can touch on and I know you're familiar with the embodiment thesis or embodied cognition I think that's the other part to Mortal computation that's important is that the body offloads some of this cognition whereas again even in the the Chinese room version of Chad GPT or whatever uh we're not necessarily using uh nonb sorry bodily non-neural like components to sort of offload this cognition so that separation that that uh blending and that combination is not there in those types of systems so perhaps that's where I think maybe I just want to remind that we The Immortal computation really refers to that entanglement right that design co-design of the software with the hardware the substrate and the uh you know the intelligent algorithm that you're designing if that makes sense nice yeah yeah that makes perfect sense and yeah it it brings in embodiment very nicely um yeah so let let's go into our triple stasis words Asis I don't know where the limit stops so I think people will be aware of what homeostasis is it's

this you know this description of the fact that uh biotic organisms have certain bounds that they have to retain you know remain within in order for their survival to be viable so you know we actually all learn this at school but you can think of glucose levels or you can think of temperature something a bit more foreign to people is allostasis which is actually quite simple it's just the idea that uh uh you cognizers capable of more uh you greater temporal planning can take prospective action to minimize future homeost you know homeostatic problems so if it if I look outside of my window and I see it snowing I can plan in advance and put on a coat before I go because I know if I go out there I'm GNA have these dis homeostatic outcomes because I'm going to freeze to death right so but you know maybe uh my dog although they love the you know actually my dogs they love the snow they don't really mind but let's say they did they might freeze to death actually they probably wouldn't but you know the bound yeah it's fuzzy who has what yeah yeah but then you introduced this third term uh you and Carl homeis thank you homeorhesis yeah to get it to say it right too wonderful homeorhesis homeorhesis yeah um let's just start with what does that mean sure um so the really simple way to think about it so that way those that are familiar with homeostasis so you could you could State homeostasis as there's a set point or a particular there's kind of flavors of it a number an average number or a range that you want to keep like your temperature again so that way we avoid leaving the non-equilibrium steady state and dying um so we want to keep things within a range and we want to keep things relatively constant right that's what homeostasis ultimately says um and then you built on it allostasis is predictive homeostasis right it's trying to serve homeostasis to prevent that hormesis one way you could take it is it's time varying homeostasis so that's the the first place you could move to which is what if that set point was changing with time right it was dynamical so in that sense that that almost gets you roughly to hormesis a better way to so that that's one way to build on the audience's knowledge another way to state hormesis is that instead of it being like you or a process returns to a set point or constant value or within a range you are returning to a trajectory so that's it's where it's like it's a random dynamical systems view of homeostasis whereas homeostasis is talking about con a return to a constant Point here we are returning to a trajectory so this that's the time Evolution and I think that to me made more sense so I'd say that would be hormesis and then by the way a fun variation to it and again you won't find these too common around is homeo hetic if you want to say something is homeostatic you would also say homeo hetic or a homeo process try saying that a few times that's yes I think I think Cole just proves as the dictionary and just manages to it is amazing it is amazing um okay that's

really interesting because so again this is this is kind of converging with a LW the reading that I've been doing I've been TR ring back through Andy Clark's work um and he has a wonderful wonderful paper called knitting your own Markov blanket yeah yeah yeah which is fantastic and he talks about how and I think this is a reasonable uh critique that was leveraged at the free energy principle and act of inference in its original Inception is that it kind of took the mark of blanket as a fixed uh statistical entity let's say um and therefore the particle the internal States and the external States it didn't really allow for you know dynamical the the unfolding of dynamic what dynamical system theory predicts which are trajectories as you said so uh Andy Clark says well let's take less of a substantial ontology and think more of process ontologies so in the same way as a butterfly you know is not the same thing through the course of its lifespan we can still identify the butterfly through a process ontology now there's a really interesting part of your paper that you know this actually I'm really glad that we we touched on this because you say a morphogenetic system from a cybernetics point of view is one that maintains its continuity and integrity by altering essential aspects of its organization and or structure and I thought this was just like really interesting because no one I think could have expected the word altering there right they would have thought that continuity and integrity is going to be the maintenance of its organization and structure so I written here at what point does a morph genetic system change its identity and what is the distinction between identity and organization so we can because we can have as you say on the Arc of a lifespan on the trajectory we have different points in which we can say self-organization according to this statistical Dynamics is happening but 10 years later there's a whole different scope of self-organization but it's still self-organizing that process is still going on but we don't want to call these these two distinct entities right the caterpillar and the Butterfly or the baby and the adult because there seems to be some Integrity or continuity to use your language is that Just an Illusion right um you know I I I come from this from uh The Works of people like Derek parfett who say that well there is no single substantial self it's psychological continuity but in a sense you can almost bracket that psychological continuity off again we're talking about lens of analysis and say okay well that's the system yeah that's the self right that's what that's what matters in the same way you can say okay we're no longer really concerned about the difference between the uh caterpillar and the Butterfly what we're talking about is the fact that butterfly follow because of genetic reasons this path and we can identify the path now as what a butterfly caterpillar is so all of that to say how do you resolve this tension between identity and organization because it seems to capture reality we really do have to integrate this

dynamism and this in this these flows over time and these changes wonderful example um the butterfly in the caterpillar I think it's Andy's um yeah I've read that somewhere too I Know Carl refers to it in one of his works as well it's probably Andy I don't I don't know um so there's a lot to imp hack there but the one concept I'll just say to start before I forget it is that the butterfly and the caterpillar example you'd say well okay are they different right you know I consider that and this is where again since we did start earlier with hormesis which was returning to a trajectory rather than returning to a constant value you can sort of borrow that idea of time evolving systems and just saying that uh the fact that that butterfly is a projection or a continuation of that caterpillar's identity the fact that Evolution that change is the the uh caterpillar caterpillar butterfly system yeah yeah yeah that is the one object and um so that's kind of like that that Evolution and I do think I touch on it in here in in some way or another about that that changes you right that that Evol because again because we are ohic your your uh the variables and the right critical ranges and things will change with time as well you'd say well wait aren't you aren't all organisms defined by this very specific set point the idea is that you have to remember you are an open system and you live in an environment that itself changes with time so the idea is that because that is evolving as well and and also it's changing because of you you're you remember we talked earlier about Niche Construction uh we are acting and changing our environment so we need to constantly be changing um so that evolutionary process I think of as is your identity and I I could have swore it might have been Andy or Carl in reference to Andy or someone that was referring to his work about that there you can look at metamorphosis as still that that is part of your it's also part of your identity preservation because it's like an adaptation right an adaptivity to a Time evolving universe um you also asked earlier so to make sure I connects so that's the the the simple answer is that I still see the the butterfly as the caterpillar it's just not called a caterpillar it's a it's the identity is still there but the structural form has been right in part of that trajectory that so the cat and again the we could also say caterpillars generally always end up metamorphing into a butterfly there is a trajectory so that is distinct that is that particular insect right right same thing with a human has a different trajectory in same you you could use your favorite animals like your dogs you mention your dog earlier you know was a puppy right puppy the same as the full grown dog well there was a trajectory it was the growth growth is part of the Mortal computation framework and we did make sure to put that in there uh and there's different ways to characterize growth regeneration and Decay uh in death um uh but you said earlier about I identity and organization um so I want to be a little careful because there's also the world of I

am familiar with philosophy of mind but I don't want to get too much in there and start angering philosophers we've already probably angered a little bit with our ignorance on physics so not try to step on too many toes but so the way I thought about it in Mortal computation is there's this idea of organization and structure you did bring up morphism so this will help connect all those parts for the audience that heard these words um and I think of them as sort of two different complementary things uh and then I'll tell you where identity comes into this picture so uh organization in from a self-organizing dynamical systems kind of cybernetics perspective is the relation the parts and the relationships between them right and so uh a human might say you could say well okay we have two arms and two legs and we are organized according to this way our cells are arranged in a you know have certain relationships to each other structure is the instantiation of the organization it is that specific you darus are very different than me uh in your instantiation in this universe right uh you and I are both humans right we both have rough organization of Parts um but our physical embodiment our actual you know instantiation into this moment in time uh is that organization expressed right in a biological form but again you could talk about as in silico entities as well and so the nice thing about thinking about organization and this is where again like that that idea of like uh organizational closure we use that in our definition as well as long as you have sort of these arrangement of parts and these relationships between them right and it's very abstract it's kind of like it doesn't want to tell you exactly what you need to create to create this for example animal um that's the identity right it is these rough relationships so that's where I want to I I don't particularly divorce the idea of the identity that organiz rough organization of parts and I say rough because you know again we can we think of it as an evolving trajectory as well the structure itself is different and that's what's kind of nice is that changes on a a faster time scale if anything else so the fact that you and I started as babies and we are now adults right that is because cells are growing cells are being born cells are dying there's also pre-programmed cell or sorry programmed cell death potosisd from the environment and it's also saying okay well I want to repair that damage and that's where the morphogenetic morphogenesis kind of Concepts come into play is you and I I think a better example you can even think about neurons in the brain uh neurons you know live for a certain time scale synapses live on a much less time scale many synapses die and within so many you know minutes or seconds and cells die every but then they're grown and reborn so you and I are no longer the same exact collection of cells that we were a few days ago we are different uh physical embodiment yet you're still darus your identity is still roughly there even if maybe

you have a few more cells right and more you you're growing and adapting to your environment so I think that's an important kind of concept is to think about the difference between organization and structure they're not actually the same thing which again if you don't not familiar with that distinction you might assume well aren't I the same structure uh no you're not that's kind of like the the actual component the physical substrate itself is in a constant process of death and birth and repair and damage and uh even the processes themselves and at some point or another in the paper we talk about that even the processes of repair are repairing themselves or decaying so there's this Con so this is where again although all that blabbing that I just did goes back to again that trajectory right you are a trajectory rather than a fixed set of things that need to maintain them exactly as they were because you're dissolution into the environment is that trajectory no longer being there right that you're not able to actually maintain your homeostasis which is returning to that and I like homeostasis is such a great word because it really just reflects the idea of what we said earlier the non-equilibrium steady state right we always want to return to that but that state right whatever that might be to uh ensure your continued organization your continued identity is changing with time right because the conditions the environment are changing you are in a naturally evolving system just by basis of you yourself acting on it so everything is constantly evolving so I guess there might be a tension I have and perhaps I entered the the the chat entered the discussion late enough that uh I don't see I don't see a problem with the metamorphosis I see that as that is the fact that you are your you are a trajectory and so yeah entity still continues it's just your physical structure might be different right and that explains growth um there is one final tiny point I'll add to it so uh and I got to be careful where the where I use the word so when we were talking about the cybernetics it's funny how biophysics and cybernetics they sort of have similar word different words for the same kind of concept uh I think that uh yeah that that's a almost tological kind of way of you know viewing it but um one thing that they have in cybernetics is called self-replication uh which is what we used in the cybernetics back end of the Mortal computer in this paper to explain that uh you know systems will uh produce more of those parts they organize and you can think of them as cells growing or dying um and if you add mutation to the self-replication you get reproduction so that is where you get this concept of okay well what if for example you have Offspring you know one day you know you have a daughter or a son um and they they are a continuance also of your identity in a way right because they are still subject to all the conditions that you were subject to right they still need to find their non-equilibrium steady state they need to preserve their organizational closure

their identity uh with an evolving structure or Morphological instantiation right but in a way your identity is carried on through them and then Carl had this really nice uh uh it was in again one of those interviews he had in the past of that it's also a way of getting what's called optimal basan forgetting where as you get older you get more set in your ways you become wide but you're less able to change and so that's the the cycle part The evolutionary Evolution's kind of answer to saying that okay you are not adapting as well as you could or as effectively because the optimal thing would be is you say okay let me start over let me and this would be related to like your neural information but you know let me start over again well you kind of can't right you you have continuously self-evidence to a point that uh things are more easily predictable for your internal generative model which is you know one way of looking at your your identity uh and you you are now kind of rigid you're less likely to change you have certain charistics that you've established right so that's why you have Offspring and then you have you know your the Next Generation and they don't have the same constraints they're more flexible they're more plastic they're more able to change whereas you were less able to however your identity still continues it might not continue directly obviously you know if you have let's say a sun not EX they're not darus but they have elements of you in there right you know the things that you interacted with them the knowledge that you imparted with them so you could even think of I'm I'm going a little a little beyond the rails here of the paper but we do touch on just like reproduction is just the mutated self-replication concept and you know once you have that you know that's how you preserve your identity and project it even further into the future in a somewhat more abstract way um I also think and at risk of blabbing any further I'll I'll stop here is that it connects nicely to one of the other motivations towards the beginning of the paper uh we only briefly and I don't want that to be kind of like a sticking point touch on uh the philosophy of life and death or existentialism which is a lot broader and it touches on a lot of other bigger Concepts than I think mortal computation but we we borrowed this idea the fact that and you said it earlier as well the fact that oh you because you brought up haiger and being in t uh the fact that we know that there is an end uh and that we project towards the end we sort of act because of that and that's where the I connect it back to the reproduction we impart information we teach right we express ourselves through music or art um I think it's a nice kind of cultural connection uh from the idea of even mortal computation which is that that's how you pass on your knowledge you're encoding your information in your Offspring and giving them somewhat of a prior right and then of course they'll obviously adapt and you know can you do their self- evidencing and modify their

posterior according to the data they acquire from the environment but that's your way your identity still carries on in perhaps an indirect way but that's a projection far beyond your own physical end in death so I thought that was a nice thing we did is that we touched on that because there's naturalist philosophy which is I think a little bit closer to you know what motivates the idea of moral computation but that existentialism is kind of there under the hood and you sort of realize well okay that's why we act the way we do we know that this isn't going to continue forever which stands in stark contrast to the artificial intelligence systems that we have in today which in principle they don't have a known end right they could theoretically keep running being copied on to a different sub so I think I'll stop or at least yeah or at least they wouldn't know right um I mean um yeah I mean a whole a whole branch of of therapy and philosophy Terror management Theory erms Becka denial of death is exact points exactly to what you're saying which is that you we are motivated by finality whether that's as strong as ear specka that everything we do is governed by the Spectre of death you know we create works of art for a legacy we have children so on I I like the hiigaran view which is that there's this very sort of subtle um future orientation in all of our actions it doesn't have to be as concrete as creating a masterpiece um but it's just the fact that and this actually might point to the incapacity for presentness in some sense is that we always have to be thinking even if it's implicit right because self-organization is always for t plus one because T 's already been taken care of you're there yeah right um yeah yeah it's all really cool I love that there's some really amazing work that's being done on on existentialist philosophy and active inference on hiar and hle and stuff yeah yeah there was a couple things that came to mind that I think is worth contributing I think one is uh a notion of philosophical vagueness that's often forgotten which is that some questions are just open right at what point system a becomes system b or or what point it retains its Aus yeah Derek parfit who again I mentioned is really influential he just says it's an open question there is no answer right you know it's it's uh feus a ship or it's sores Paradox at what point does the grain of sand the accumulation of the grain of sand become the Heap well depends on who you ask and that's fine I mean it might just be our our proclivity to have um binary results right we just want a yes or a no the other thing that came to my mind was that yet you know we're talking a lot here about change I guess the one thing that remains is the the mechanism to be able to say that you are a thing has to be the same now whether you call that self- evidencing or self-organization or the fact that you are matching operational closure with thermodynamic openness or sense making whether you're the whether you're the caterpillar or the butterfly you're still abiding by those laws and so I think that's where Carl's

stressing that the free energy principle doesn't tell you what's going to be there it doesn't tell you what's going to be there now or in the past or in the future it might come to that because I know they're doing things about scaling up Markov blankets and and now we're having Dynamic Mark of blankets and we Mark of blankets but it's very raw form it just tells you if something is here now and if it's going to be there at t plus one and we're going to say there the same thing axiomatically they must be minimizing free energy just on pure quite boring maths and so it made me think that like the but all of this stuff is hinges on the fact that there is some level of self-evidencing going on so the mechanism itself stays the same yes so I thought that was yeah yeah so I thought that was interesting all right cool and you're still and you're still interacting with your environment through your Markov blanket the the morphological substrate instantiation of it evolves with time you're still there it's still a mark blanket you're still interacting with sensory and active States yeah yeah yeah exactly you're sense making but in the what's what's cool about sort of Evan Thompson's work and the inactive uh tradition is that the the autopoiesis is is is also so it's not just that sense making is happening but also Autopoiesis is happening it just depends on your scale analysis right so you have cell autopoiesis as well as cell sense making but that's embedded it's not it's not distinct it's embedded within the autopoietic sense making of a nervous system which itself is doing sense making because it's got act it's got sensory motor uh relations going on yeah so it's not just that this is you know transformative of x to Y it's also it's also recursive m mm that you could have like a and b bracketed within a or what you know whatever like it's it's like language in many ways right we've just got loads of relative clauses and you know these are embedded Markoff blankets um we also do that in the paper we also have a I I don't know if you caught that uh uh mortal computers within mortal computers we had yeah yeah yeah C subunit because I did talk about it in yes yeah yeah yeah yeah and they one mortal computer sets the set point of another mortal computer computer so you get that recursive beautiful construction that you just mentioned there you can have that in the framework as well I like that yeah yeah yeah it's wonderful um so I wanted to ask one more kind of specific technical thing about the paper and then a couple final more you know more general questions sure I thought this notion I'd never come across before and I was really I really liked it was cybernetic variety ah I thought this was really cool and the law of requisite variety I thought these were really really fascinating um additions to the paper by the way for people wondering because you know this is in the paper this is in the paper the paper is big and bold and it's got a lot in it and it has appendix too and it has an appendix which I didn't read um you didn't need

you didn't need to I no the paper's fantastic but yeah it has a lot in it and I wanted to and I really enjoyed this bit on cybernetic variety so maybe just for our audience if you could explain what that means me sure uh and I remember too reading that there was even in cybernetics confusion as to what variety was but uh I think the simplest way to look at it is that it's the number of states in a in an entire system so the cardinality of the state space so uh in very specifically it's actually the log base two of the cardinality of the state space to give it so uh we got to give uh so the thing I we got to give Ross Ashby who's the the author of things like the law of requisite variety and the good regulator theorem actually one of the one of the many uh core contributors to the field of cybernetics we we gotta talk about Wiener and the other wonderful guys that were cyberneticist as well but um Ross Ashby made some really wonderful contributions and so variety effectively is like I said the different number of different states and I was thinking about this last night in preparation for this meeting just to remind myself so one thing you can think of it is that you have the number of distinct kind of configurations and you could think of um I'll use an example I think it's better to explain it let's say you're trying to debug um the fact that your light's not turning on so you have one component you could look at which is the light switch and so it had light switch itself has two states on or off so it's binary right so the variety of that's actually log base two of two which is one so there's one difference or distinction and that's one re that's where the word variety I believe really gets um but no one really quite says it except Ross Ashby in his textbook you can find it somewhat described in there um there's one distinction between the on and the off SE and that's by the way that is variety right there and it's wholesale but one way to think about you're trying to debug why your light's not turning on so you you identify oh the problem is in the light switch um so this will help us understand the law of requisite variety you had uh you know one controllable changeable aspect of the environment which was one thing um and you had one hypothesis or one kind of uh thought internally about how to fix that and so one you're you had internal variety of one and the environment ah yes I see where you're going nice right just to help us connect together variety is useful to understand the cybernetic principles um okay so that's great law the variety internal variety was One external variety was one now let's say that it wasn't the problem the light switch is not the reason why it's not broken so now you have to figure out something else in the system so there's also uh let's say I don't know a battery or an internal component that's really I'm not going to give us the right Electronics engineering here but there's another component that you figure out ah it had an it's either broken or not broken right it had these two possible States or like let's say it's

um sorry it's either this or it's that you figure out that was the source of the problem well your variety is two because it's a sum of these independent components that make up the system so it's you had two binary States and you just sum up these independent components that are not necessarily we're going to act as if they are directly uh jointly distributed they are independent components you solve the problem your internal variety was two the external variety was two you solve the problem because solving the problem by the way is that you found stability right you you fix the issue that you don't want to be in a dark room right that's kind of against your life right that's sort of like you want you want the room to be lit up so that was the other thing and so then you could continue extending this out and finding chains of things that are problematic and you keep summing up the log base two of the number of distinct states that these components have but let's say that you just did those two and that wasn't the problem and that's it those are the only two things you can think of from your internal model of the environment you're like well there's a light switch and I know there's a power source to that light switch I've looked at both those things they're not broken I fix I I still have a dark room well th the environment has greater variety than you it has it turns out that if you had known there's actually a power breaker outside all you had to do was like flip it on and off and you had on but you never you never get to that so the external variety is three but your internal variety is two you have now failed or you you are violating the law of requisite Variety in the sense that you are not going to be able to solve your problem you won't survive and you could then connect this problem solving back to homeostasis or hesis where you're trying to block or protect these life critical variables or metabolic variables which Ashby does talk about them he doesn't maybe always call them uh meta metabolic things but he does use the word homeostasis which is another way of saying in cybernetics Ultra stability which that you're able to very quickly go back to that optimal configuration or that good configuration you have currently found and so so that's like the law of requisite variety and helping you define variety all at once I think it's good to just connect the two variety is useful though for like a ton of other principles and I think you probably you mentioned law of requisit variety and the good regulator theorem uh there's a lot of other and there's even more there's a few others that I didn't include in the paper um about they're all driven by variety like for example blind selection and retention uh and order through noise because this touches on I noticed uh chaotic systems and random dynamical systems there's a lot of relationships and overlap between modern-day cybernetics uh and those principles but like the idea in terms of variety is that there are all these states in the environment and what you want to do is you want

to try to reduce that variety you want to find configurations where there is less variety because then that will be your ultimate goal and we uh Carl and I sort of reframed a little bit cybernetics it were reorganized it into these pillars so one of them is what's your goal which is you want Ultra stability not just stability Ultra stability which is their your ease in which you can self-organize or get back into that condition where again you're internal variety is balanced to the external variety you're able to continue your homeostasis you not going to dissolve or disintegrate um the other one that we had was uh what was it uh regulation or was it no it was oh it was growth sorry I gotta remember there were three one is growth of course the idea and cybernetics does talk about this is where I said earlier about the principle of self-replication there are two other principles I just want to tag in there the principle of autocatalytic growth and the principle of recursive systems which is that um you know the idea is that once you start to find the right resources right and assuming that there are enough you're going to grow and grow bigger and bigger and bigger because the idea is that you want to preserve your configuration you want to continue um and then recursive systems just talks about the fact that well these little building blocks as they you know stabilize into configurations and replicate and you know you get more of them they start to connect to each other and this is how you get complex systems right you can build heterarchical or hierarchical systems and this gets into Herbert Simon he wrote some nice papers about this so you get more and this is you could relate that to cells like for example single cells organizing and combining together create multicellular structures which eventually give rise to organs which give rise to subsystems and gives rise to you know complex entities like animals right yeah and then the final part is regulation which we just talked about uh law of requisite variety uh and I should mentioned to like you you want noise in this process this is the other one of the other pillars that helps us search through the space this gets a little bit like reinforcement learning for exploration you need noise and then that's how you get order it's called the signal to order uh principle and then the final component is like you said law of requisite variety the good regulator theorem I don't know if you wanted to make the audience clear on that but it just basically says um in order to be a controller of your environment or something that controls its environment is a model of its environment refers to its internal generative model and so in order to be good at manipulating and doing Niche construction right you need to have a good internal generative model and what better generative model than to be a model of your environment or the world you live in and then there's one last one is called model of internal model control or IMC principle uh I just wanted to give that guy uh that principle credit it's not fair sometimes you

see like oh it's only Ash beads like no there are a lot of other wonderful yeah yeah scientists there and that just basically talks about U we talk about a synthesis of a controller in environment and we want to deem it to be uh stable from what's called a closed loop point of view um because that's where feedback is introduced and so there's this basically just an extension of like the good regulator theorem uh I'm trying to remember his name this is what makes me feel bad because I wanted to highlight his principle but he worked actually with ashb on the good well good regulator is Conan and ashb yeah Conan that's the other name I wanted and he had another principle the model principle internal model control which just talked about uh you want uh stab you want it to be structurally stable from a closed loop perspective and that that's another component that you would like in your system so I think with all that that gives you your definition of variety that's that's nice I gotta get more into my cybernetics um wonderful yeah yeah really cool it may there there are several things that popped into my head there I mean one is I think there's a really nice uh mapping between the law of requisite variety and basian inference in the sense that to be able to conduct a posterior I need my States and I need to conduct okay what's the probability of the state given an observation but that itself is making the inference that that state is there in the world so I have to have a kind of isomorphism between the states in my generative model and the states I know or or My expected States and the states are in the world that said something that I was thinking is going back to our conversation about entropy and again I'm not going to annoy the the physicists but as I reduce my internal variety I'm bringing entropy down which thermodynamically means that I have to be introducing some variety in the environment yeah yeah so I think what's interesting there is I think it's just in the term and I think it's very clever term law of requisite variety it's not like the law of identical variety because I have to have enough that I can do uh you know sufficient tracking of the environment but that doesn't mean that um like that that is like sufficient that I have enough but I you where am I going with this you know where I'm going with this um I was F yeah yeah yeah I W I lost myself but in the sense that yeah as I bring it doesn't mean that the the external system and the internal system are always the same because it just depends on sort of your again everything comes back to the kind of lens of analysis you might be causing entropy elsewhere but you just need enough variety so that you can account for your little niche over here exactly I think that's I got there I got there I got that no I was F I thought that's where you were going because it was talking about like your local environment where you right exactly because you're not all over the universe exist in this as we said earlier in our least offensive way the entire universe is the

complete closed system so that's what allows us to uh be happy and uh and reconcile the second law of Thermodynamics but you and I while we are in the entire universe we actually really care about and we are working and operating in this very local component of the right right local Eco we're not everywhere on the planet because you know that's we could talk about the collective but just about you and I think that's what you were where I think you were going and that is your law of requisite righty really respond is really about your local Eco Niche related to you at this moment en time yeah exactly exactly we're not talking about the variety of the entire universe precisely precisely so yeah I guess the optimal thing is just to dump my entropy elsewhere and just run back to my bed you know exactly throw that inop away yeah yeah throw that in away okay cool that's really good I just wanted to uh ask I guess a very theoretical question which I think is a kind of yes or no which is that if so with your I thought your light bulb example was really really useful if let's say uh so there are multi so in terms of the external variety let's say there are multiple binary things that might be causing the light bulb to not be working so the switch the battery the fuse whatever but I get it right the first time but I've only got the knowledge of the thing that I got right the first time so let's say it's I I go for the battery and I think oh I don't know anything about light bulbs I'm four years old I only know about batteries let me go look at the B oh it was the battery was dead right has the four-year-old in that example like what what are they intelligent it's a kind of um it's a kind of GTI problem I'm not sure whether you're aware of Geo problems Geo was a edman g was a French philosopher I believe who basically since Plato knowledge was defined as Justified true belief so um how do I know it's raining well I can look outside and I believe that it's raining it's true that it's raining and I have a Justified belief it's raining because I can see that it's raining but if let's say my cursors were closed and I said and it was raining and do I know that it's raining I say it's raining I believe it's raining but I'm not justified in my belief so get so that was kind of since Plato what people thought knowledge was edman Geer comes up and basically says for example it's like the same way that a clock is right two time a wrong clock is right two times a day I can look at that clock and it might be accidentally true that it's that time but is it is Justified because I'm using a clock it's true and I believe it but do I know that it's that time because I've looked at that clock well see so that's the philosophical background um I can't really remember where I was going with this oh yeah but the the foury old who's got less who is not obeying the law requisite variety who's got minimum uh fewer sort of internal States than the actual volatility in the external world how does cybernetics deal with that individual so I mean at that moment in time right you have less your internal variety

is less than external variety So you you're in trouble but the component that I at least where I'm going for is that we're we're forgetting too that there's an evolution and time adaptation right to the system so that four at that moment failed to identify the problem but that doesn't mean that they can't explore and like you said earlier I think you gave me sort of a piece of the answer you said well they look up batteries or you looked up a battery and you learned how that works you might later learn about the circuit breaker so the LA cyber ICS you got to understand that you are in the process of reducing your variety and so you are changing your configuration so the the configuration or changing your state means that earlier principle I mentioned the the principle of uh order from noise I think is important because it always tells you we are trying to find attractors in our state space right it's an optimization problem and we want to find an attractor or a pullback attractor is what people say where it's like something you like which again connects nicely to Ness but without repeating ourselves uh you are trying to find these attractors and to reach again like I said your ultimate goals Ultra stability you want to be really good at solving this light problem you want to understand your generative model sorry you want to understand your world which requires you to update your generative model there's a learning component an evolving adaptivity process and that's where like the good regulator theorem says well you know you if you're a good controller of your environment you're a model of that environment but you might not start out that way you need to adapt you need to acquire information so exploration and exploitation is actually in cybernetics even though you know we typically nowadays think oh that's just the reinforcement learning concept no they talked about exploitation which is I like my particular tractor I'd like to stay there and it's not very easy to leave it however that noise is what pushes us out and learns to find a better configuration there is also some other principles in the law of requisite variety The Law of asymmetric transitions which says that you are very it's less likely for you to go from a lower variety area to a higher variety area um so that means we're always looking for those spaces where we have less variety and again we could use what you and I discussed earlier your local Eco Niche you want to find such that you have the least amount of intern external variety so that way you can match it right to satisfy the law of requisite variety um but the other part is the order to noise principle and then there's uh the The Selective retention principle and I'm going to mess these up even though I made sure to carefully outline all of them in the paper so I do recommend to read that to see their their interrelated uh components but the idea is that selective retention and blind uh and blind selection is the idea that you're more likely to stick to better configurations or states that you

find in your optimization process but that means you have to be changing you have to leave where you currently are especially if it's a very if it's a suboptimal configuration so I think that four-year-old that you talked about earlier will eventually I mean it might be a little difficult for you know in this practical example to figure they're going to figure out what a circuit breaker is but maybe they're looking up things on the internet they're learning things about Electronics because they're like I really don't want a dark house and they're not going to just give up let's assume that they're going to want to tend because they want to survive in the light represents the survival they're going to learn about the circuit breaker and so ultimately they are evolving to a different configuration that now they have a better generative model of their environment or it's more close to the actual environment right because this was a factor a causal factor that they just hadn't identified but once you learn about that circuit breaker now that's back into your system now you understand its amount of states and now you can match the external variety of your local Ecco Niche or the components of the local Ecco Niche that are related to the problem you're trying to solve because I think we could probably cut up the niche in different ways and say have different goals like the light only requires these pieces of the environment the other parts are not relevant to this problem so we don't have to even account like you we said earlier for that variety we only need the variety that's related to the circuit breaker and you just need to uncover through intelligent exploration which connects back now we can even say oh that justifies active inference we need to actively sample our environment so that we you can learn more about how the electronic system works so that way you can turn the lights back on does that make sense very much so it was thank you for doing the active inference spin I was going to do it um we're maximizing our basium surprise we're we're doing our K divergences between our prior and posteriors about future States but no you it's fine um but no very good exactly um it's not it can't all just be pragmatic value there are epistemic affordances that we need to not exploit but we need to access um to to be self-evidencing and that's all in the equations for people who are wondering um for that that K has derived in terms of expected free energy in this case excellent okay so we're coming up to two hours I've got a couple final questions one is very uh well you can take it as long as short as you want but I I was just interested to see I've I've taken this habit of when I read papers I love reading the reference list um because you know how they have numbers so you know you either put the names or you put the numbers and I love looking at the numbers and you have a lot of numbers in yours and I couldn't help but stumble across um Donald Hoffman uhuh I don't know if that was you or Carl um let me just look at what the title was because I did it

would have been his what's it called um oh I can't remember what the name of his paper is in any case it doesn't really matter um it's just he has a very funky theory about what reality is um and I anyone who knows about Donald Hoffman I have to ask about Donald Hoffman but let's skip that then um final question is there's a big hubub would say sorry not steal your thunder here but yeah because it says the interface theory of perception right so uh can you briefly remind me because I feel like I that'll help trigger why I included him in the referee yeah yeah so Hoffman has this idea that we are evolved there's multiple points of convergence here but basically that we're evolved not for a veridical perception but for survival and so what we actually perceive is not what's really out there but what but Fitness payoffs and what he says is so that's not particularly controversial you have that in indirect realism you know you have that going all the way back to bur and Russell and even prior but way what what becomes sort of radical about Hoffman as he says well time and space itself are part of that interface and he derives a lot of this from theories of physics where he says okay now physics modern physics I believe the no in 2022 was one for something along this area um is showing that there are these symmetries outside of space and time he has these extremely funky Greek derived names um which I won't even try and pronounce and he's very popular on um in in Academia and in popular Academia so whenever I see his name come up because I you know read his book and I'm interested in his work I I like asking people but in any case it does it doesn't the first component of what you explained is why least I remember I Know Carl and I had different references but I remember that resonating with the survival component of the argument that's probably the piece of his work that we thought resonated in this work not the other I don't have anything to say about the other part although I should dig a little bit further and see if there might be something that changes but well there he's got plenty of podcasts out there and his book is very good what's it called anyway you you know his name on go it's all um so the final thing I wanted to ask is a kind of more broad sociological thing which is that there's a huge hubbub going on at the moment about AGI artificial general intelligence and the fact that you know Sam Alman came out and said that um neuronet or large language models are not going to get us there and I think people have thought that people have had their suspicions since the 90s since inactive uh you know in activist cognition for E cognition because where are you action perception Loops right whereas your grounding in the world and so on and so then we have a company like versus which is trying to implement active inference uh intelligence into you know in silico just as a computer scientist I can't help but ask you you know what do you think is going to be the trajectory um and do you think mortal computation

in the form of active inference is going to get us there oh this is a this is a good one and I I feel like I should be a little careful with my answer because there's that not with you but there there's politics in yeah about what we think might be the future um uh so I guess I'll say one thing about the paper because the paper does sort of touch on this right it actually has a section towards the end that talks about um the need to reconsider the path of machine intelligence I think you may remember we talk so Carl and I mention in there uh about tool Centric AI um so in that regard just from the broader perspective without I I'll decouple the versus thing in just a minute and I'll address the active inference component um while I don't know if mortal computation is the only path I I we do say it's one possible viable stepping stone to a different path and I do think it's a a Recon consideration of how we build intelligence systems and I think that that that's kind of it's another so even though it's a survey right because it's trying to survey all these wonderful ideas from brilliant thinkers Engineers uh theoreticians and engineers and scientists over across all these different domains um that we're trying to say this Echoes a direction that we want to build our our adaptive systems right and we don't have that right now I mean we might we do have the what do you call flickers of it and uh the other piece I just want to point to the audience is towards the end we give examples of what we would consider viable mortal computers that are not even necessarily all in silico Ross Ashby does deserve a little more love and attention he has the homeostat which we actually have a little pseudo taxonomy and we call that a homeostatic mortal computer although it can kind of be allostatic depending on uh the later uh Innovations other scientists made on his homeostat um but we also talk about these biotic uh mortal computers like fungal systems we talk about uh organoids which even though I haven't had the pleasure to work with them I'm deeply fascinated by or the sentient organoid as uh Carl has called them in snova and those that have done this wonderful microphysiological construction of these little entities that can play Atari even if it's for a short time span it's still fascinating to me uh that we can use stem cells to do this um also we got to get Michael Le some wonderful attention we celebrate the xenobot and the xenobot Swarm which can do these wonderful things so I think those are examples of the type of intelligence that we would want to strive to emphasize and build more like because they can do things like self-repair and self-damaging they are truly adaptive in the sense that I don't think the robotics that we have today can do right so things are very brittle it's very hard you want to I I tell some of my colleagues you want to sober up about the state of AI because you know there's a lot of wonderful things that current models and systems can do uh but there's a lot that they can't do I say go to a robotics conference and you'll see instantly how far

we are from AGI or even just really powerful AI narrow AI like to get a robot to get you a cup of coffee although there's some nice recent advancements in robotics but we still have a long ways to go I think to get and those are the types of problems that I think emphasize where mortal computation might play an important role it really gets into into this other domain of Robotics that I've been fascinated with which is neuro Robotics and neur Robotics says very shortly I try to distill it is um we want to build an intelligence system like a like a brain If you will something that's biological within the robot it must learn and do inference and engage with this environment inside the body of the robot there is no simulation uh you must have physical Hardware to do this otherwise it's not neur robotics research you're just doing simulated stuff as I have done um and so I think mortal computation uh sort of emphasizes and celebrates those domains that really require this entanglement right we want our neural adaptive systems to actually work in noisy uncertain environments and the tools that we have today with back propagation of errors and reinforcement learning and modern-day deep neural Nets they're not always the best they're not best suited sometimes to these particular problems so I think that that's me waxing about the general problem uh I will comment that while Transformers and the models that we have today are doing some really wonderful things and there's no dismissing of that uh I do not believe that the the uh key to artificial general intelligence lies in that they are very still tool oriented and while that's important in stands to make wonderful impacts and health care and other aspects of humanity that's not going to get us to the dreams of original AI right to build cognitively viable functional General intelligent systems that can act even like I'd say even at the level of a simple animal forget about humanoid intelligence so I do think moral computation says maybe we should consider these Concepts and here's we even give a definition in the paper it's not perhaps the most perfect definition and it'll probably change if more people work on this um but it gives you some ideas about how origins of Life tie into mind and how maybe what are the design con what are the design principles we should be considering to actually build this because another under the hood of my lab coat if you will is I'm an engineer so uh and uh someone that I should mention did some light reviewing and nice feedback for me not light reviewing reviewing was my Father which you might see in the acknowledgements Alexander Ria senior so just it's not me thaning myself it's me thinking my dad um and he had told me in one of the very earliest drafts of this he says you should probably emphasize how do you build this I mean obviously we don't give you the final answer but that's where the engineer ing takes are in there where I try to explain well okay this is where the sensory motor dependencies might exist on metabolism

homeostasis allostasis and you notice that so that's where to encourage the audience read the paper because we talk about where autop pieces fits in this framework and PES so I think that's kind of one General answer about AGI and I don't think the current path we have will get us there uh I could be wrong Carl and I could be wrong uh now to address versus and uh versus Ai and active inference I do believe active inference is very promising uh again it might not be the complete story it might not be all that we need um and I do think that it is but it is important that we make advancements in active inference I think that uh you were you said very early in this interview that you discussion that you were talking about you were gonna mention reinforcement learning right so may I can help bring that little tiny sprinkle in here and say that um I think it's important that active INF seems to get a little overshadowed by reinforcement learning by if you can design a really good reward function uh you can you know build a sufficiently operating system and we're good enough uh but they're very brittle they don't work very well in the continual learning setting they kind of dropic forget it's very hard to train them online they're sample inefficient I could go on and on about all the problems with RL and I think active inference STS to say wait a minute maybe we should turn this problem on its head and you know as long as we have some definition of a prior preference which I think is one of the more difficult components of active inference to get right but that's the uh inductive bias that Evolution would give us so at least it's got the connection to uh biology uh active inference really emphasizes that intelligent exploration and there's some nice results that I've seen out there that you know reduce the number of samples that you need to learn a good policy so I think active inference represents a really nice alternative uh to reinforcement learning and it even has some really nice bio neurobiological and Cognitive Connection so it's gives you that like process Theory it's often called a neurobiological process theory for adaptation so I do think that's important and I'm happy that uh versus AI uh is uh pushing along that path and uh obviously I might be a little biased because I work with Carl I work with Tomaso salvator he's also there and Chris Buckley uh so even though I'm not part of versus Ai and I don't speak for them in any capacity um obviously I'm going to be biased because I collaborate with a lot of the people that are there um so yeah I thought that was very diplomatic uh watching there's no no saying who's wrong or right I but I do urge I do want to say I urge everyone to consider mortal computation and crabs fleshing that out further because I think there's still this is a stepping stone paper I do not by any means think we said oh just do what we say it's like to get you thinking right yeah yeah yeah well hopefully you know in addition to the paper this has got a lot of people thinking um but it's got me thinking I'm gonna

and I'm going to go and practice the pronunciation of homei homeorhesis homeorhesis and apparently also Laboratory um I didn't know that one was a problem um Alex thank you so much um it's been an absolute Joy um just for people wanting to know more about you what you know what work have you got coming up and if people have got questions can they reach out to you how should they reach out to you please uh uh well first of all I want to thank you for having me this was discussion I very much enjoyed talking with you about Mortal computation and AGI uh and even poking at the holes in my knowledge there was a few points there I was like oh no I'm not a physicist and I should be clear here by the way so this was wonderful um in terms of uh how to reach me I mean there's uh my email contact I'm also on LinkedIn apparently that's becoming sort of more the social science platform I've noticed as opposed to Twitter X as they call it nowadays um so uh you can look me up there and I do now start to engage with information that people are posting there and I make announ I just made my first announcement uh on there about one work that I would like to highlight here so while uh Carl and I did this wonderful uh survey on Mortal computation and you know kind of provided definition another massive survey of mine so if you thought this was long I have a longer one uh came out at the end of December it's a a called a brain inspired machine intelligence a survey of neurobiological credit assignment where I also review over the last several decades as much as I possibly could I know for a fact I probably still missed something uh all the different approaches that have been uh neurobiologically inspired to training neural networks or constructing new neural models that are more biologically faithful and just don't do it with back propop so basically everything that's wrong with backprop and we look at everything ranging from like boltzman machines to recirculant autoencoders all the way up to Modern Day stuff including uh Jeff's work on Ford forward only learning and some other flavors of signal propagation nice things so I I really really urge that one I like to pull uh plug that work because I also think it's a nice resource for uh PhD students and posts who want to maybe engage in uh uh alternatives to the modern days we do Ai and I think that gives you a lot of uh uh uh historical context and it makes you pay proper credit assignment to the wonderful Minds so many researchers over the decades you know sometimes get overlooked in this uh you know rapidly mov Information Age so that's one paper there is one more survey that I also Rec so last year was the year of surveys for me another survey on predictive coding I know you and I didn't really touch a whole lot on it but um Thomas o Salvatore is my friend and he and I uh sort of and Anor mly we set up this wonderful survey Carl's on there Rish Ral one one of the most important figures in predictive Cod Chris Buckley's on there Thomas LTZ

um I recommend people look up it's like brain inspired computational intelligence via predictive coding uh and that reviews everything that's happened in predictive coding as much as we could over the last several years for machine intelligence so I think I'd like to those two specific ones they're very Ed and well we'll put them in the we'll put them in the video description so people will have direct access to it as well as the Mortal computation paper we've been discussing today perfect and you can reach me by my email I should last point I I don't know if you want to give that in the link to the video or feel free to say now so that you know um sure ago cs. rit.edu you can reach me there cool I'm sure that's on the internet as well um everything's on the internet Alex thank you again so much um it's been really really fun and super informative so thank you for me from the Institute as